

Simple Network Management Protocol

- Topics covered:
 - SNMP Overview
 - SNMP Version
 - SNMP Messages
 - SNMP Configuration (basic)

SNMP Overview

Simple Network Management Protocol

SNMP (Simple Network Management Protocol) is an **industry-standard framework and protocol** used for network management. It was originally released in **1988** and defined through RFCs by the **IETF (Internet Engineering Task Force)**.

Although commonly called a protocol, SNMP is actually part of a **larger framework for managing networks**. Early RFCs (1065, 1066, 1067) defined SNMPv1. These provide historical context, but the key focus is understanding how SNMP works, not memorizing RFC numbers.

Despite the name “simple,” modern versions (especially SNMPv3) are more complex. However, for foundational learning, the goal is to understand:

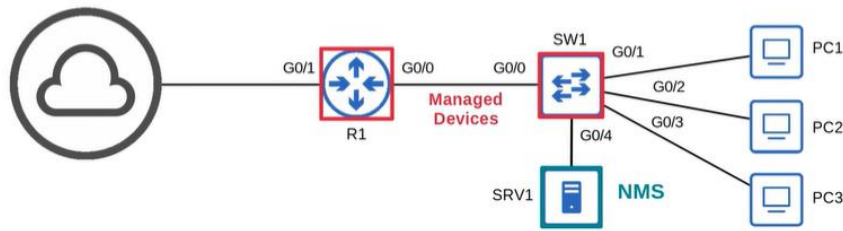
1. What SNMP does
2. How devices interact
3. How data is structured and exchanged

SNMP is used to:

- Monitor device status
- Collect performance data
- Receive alerts about failures
- Make configuration changes on devices

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SNMP Device Types (Foundation of the System)



SNMP operates using two clearly defined roles, and understanding this relationship builds the foundation for everything else.

1. Managed Devices

These are the devices being monitored and controlled by SNMP.

Examples: Routers (R1), Switches (SW1)

They contain information such as:

- Interface status (up/down)
- CPU usage
- Traffic statistics
- Device temperature

These devices do not manage themselves centrally—they are controlled and monitored by the NMS.

2. NMS (Network Management Station/System)

- Central management device (SRV1)
- Often referred to as the SNMP “server”

It is responsible for:

1. Monitoring all managed devices
2. Sending requests for information
3. Receiving notifications (alerts)
4. Sending configuration changes

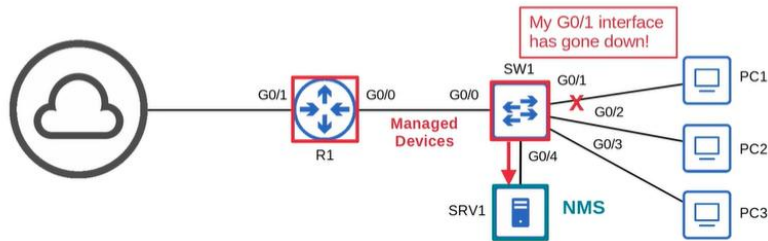
Key relationship: NMS manages → Managed Devices respond

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Three Core SNMP Operations (How SNMP Works in Practice)

These three operations define how SNMP is used in real networks.

1. Event Notification (Device → NMS)



SW1 → SRV1 "My G0/1 interface has gone DOWN"

Explanation:

- Managed devices detect events
- They send SNMP messages to the NMS

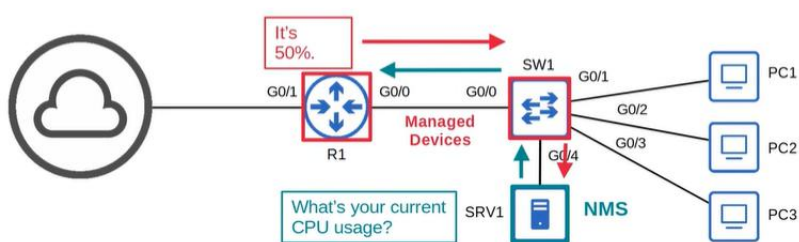
Examples of events:

- Interface goes down
- Hardware failure

This allows the NMS to:

- Alert the administrator
- Trigger monitoring or troubleshooting actions

2. Information Request (NMS → Device)



SRV1 → R1 "What is your CPU usage?"

R1 → SRV1 "It's 50%"

Explanation:

- NMS actively queries devices
- Devices respond with current status

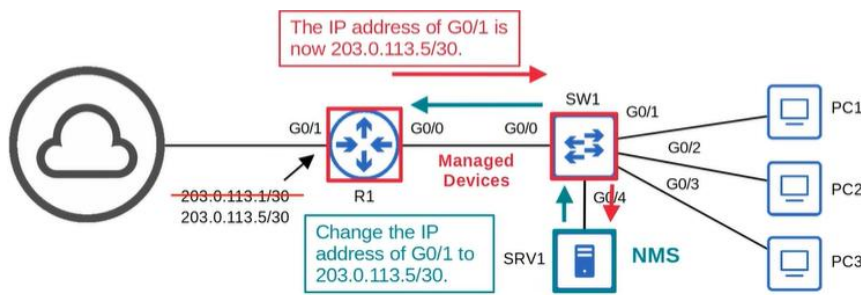
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Used to monitor:

- CPU usage
- Interface utilization
- Performance metrics

This ensures devices are not overloaded and helps maintain network health.

3. Configuration Change (NMS → Device)



SRV1 → R1 "Change G0/1 IP to 203.0.113.5/30"

R1 → SRV1 "Configuration updated"

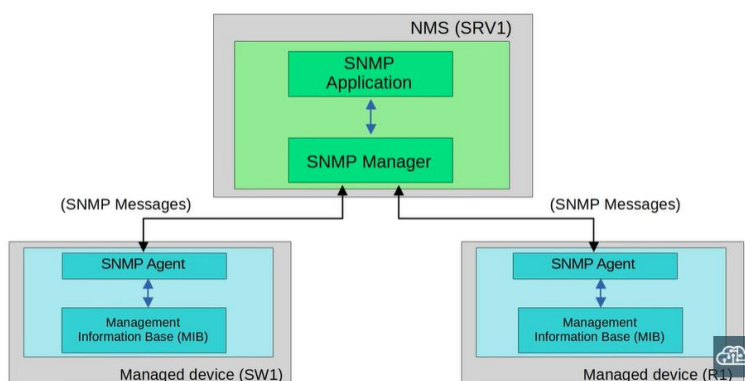
Explanation:

- NMS sends configuration instructions
- Device applies changes
- Device confirms back

Example:

- Changing IP address
- Adjusting device settings

SNMP Architecture



This architecture shows how SNMP is structured internally and how communication flows between components.

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SNMP Components (Detailed Understanding)

On the NMS (SRV1)

1. SNMP Manager

The SNMP manager is the software on the NMS that interact with the managed devices, It receives notification, send request for information, sends configuration changes etc.

Responsibilities:

- Sends requests
- Receives responses
- Receives notifications
- Sends configuration commands

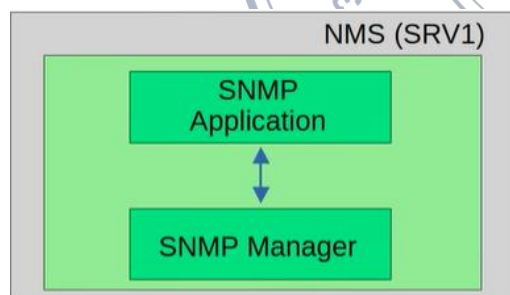
2. SNMP Application

The SNMP application provides an interface for the network admin to interact.

Displays:

- Alerts
- Graphs
- Statistics
- Device status

Important: Admin interacts with the application, not directly with the protocol



On Managed Devices (R1, SW1)

1. SNMP Agent

The SNMP Agent is the SNMP software running on the managed devices that interact with the SNMP Manager on the NMS.

Functions:

- Communicates with SNMP Manager
- Sends notifications
- Responds to requests

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2. MIB (Management Information Base)

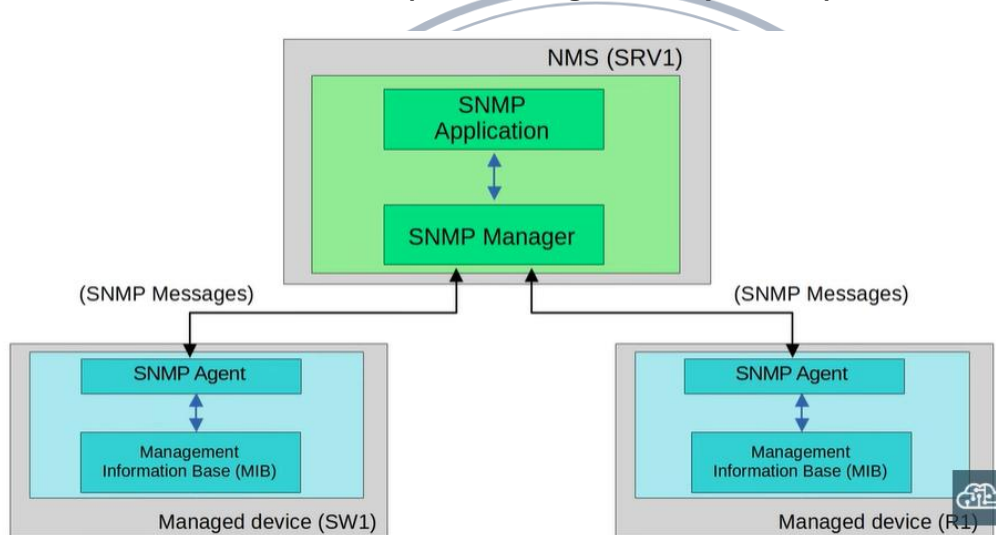
The Management Information Base is the Structured that contains the variables that are managed by SNMP.

Contains variables such as:

- Interface status
- CPU usage
- Traffic
- Temperature

Each variable is uniquely identified using an OID (Object Identifier)

SNMP Communication Flow (Connecting All Components)



Explanation:

- NMS sends a request
- Agent retrieves data from MIB
- Agent replies to NMS

OR

- Agent detects event
- Sends notification to NMS

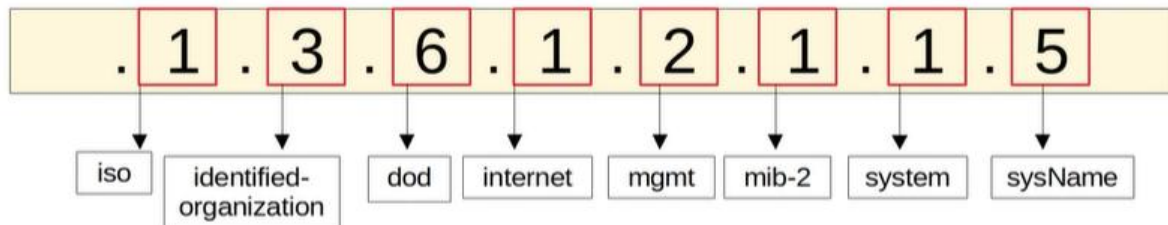
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OID (Object Identifier) & MIB Structure

SNMP data is organized using **OIDs**, which uniquely identify each variable.

SNMP object IDs are organized in a hierarchical structure.

Example OID:



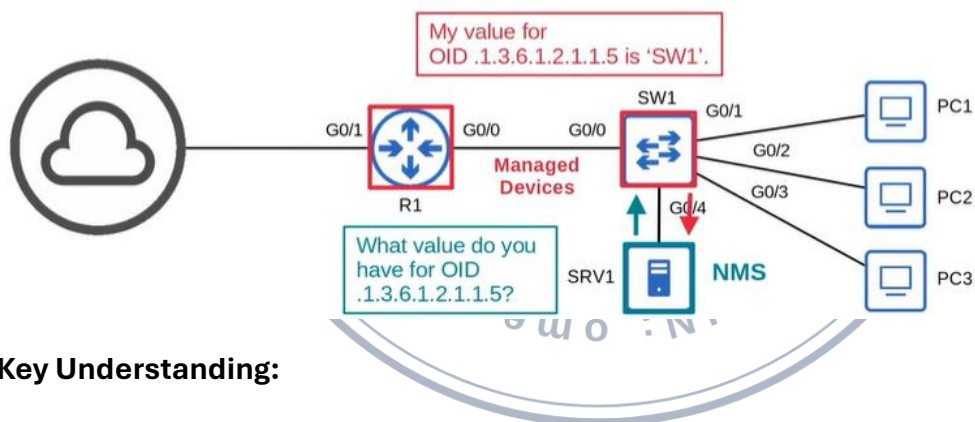
So, this OID is used to identify the system name, the host name, of the managed device.

For Example:

That Same network again, The NMS asks one of the managed devices, SW1

NMS → SW1: "What value do you have for OID 1.3.6.1.2.1.1.5?"

SW1 → NMS: "SW1"



Key Understanding:

- Each MIB variable = one OID
- OIDs are hierarchical
- There are many OIDs for different metrics

Examples include:

- Interface status
- CPU usage
- Hostname

SNMP Versions

Types of SNMP Versions

SNMP has many versions, but **only three are widely used**. Understanding their progression makes everything else easier.

1. SNMPv1 (Starting Point)

- First version of SNMP
- Basic functionality
- No improvements in efficiency or security

2. SNMPv2c (Efficiency Improvement)

- Most commonly used version in real networks
- Main improvement: better performance

Key Feature:

- GetBulk message
 - Allows retrieving large amounts of data in one request
 - Reduces unnecessary traffic

Important Detail:

- Uses community strings (passwords)
- “c” = community

→ ‘c’ refers to the ‘community strings’ used as passwords in SNMPv1, removed from SNMPv2, and then added back for SNMPv2c.

3. SNMPv3 (Security Focus)

- Most advanced version
- Adds strong security features

Provides:

- Authentication
- Encryption

Meaning:

- Only intended devices can read messages
- Prevents packet interception

SNMP Message

Message Class

Message Class	Description	Messages
Read	Messages sent by the NMS to read information from the managed devices . (ie. What's your current CPU usage %?)	<i>Get</i> <i>GetNext</i> <i>GetBulk</i>
Write	Messages sent by the NMS to change information on the managed devices . (ie. change an IP address)	<i>Set</i>
Notification	Messages sent by the managed devices to alert the NMS of a particular event. (ie. interface going down)	<i>Trap</i> <i>Inform</i>
Response	Messages sent in response to a previous message/request.	<i>Response</i>

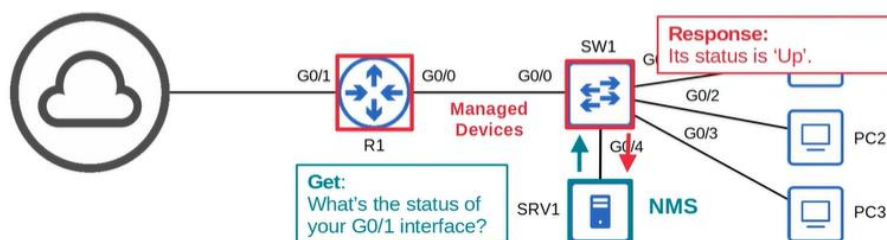
Read Messages (How NMS Gets Information)

These messages are sent by the **NMS** → **managed devices** to retrieve data.

Real Understanding Flow:

NMS → Device: "What is your CPU usage?"

Device → NMS: "Here is the value"



Types of Read Messages:

Get

→ A request sent from the manager to the agent to retrieve the value of a variable (OID), or multiple variables. The agent will send a *Response* message with the current value of each variable.

GetNext

→ A request sent from the manager to the agent to discover the available variables in the MIB.

GetBulk

→ A more efficient version of the **GetNext** message (introduced in SNMPv2).

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Write Message (Set – Changing Device State)

Used by NMS to **modify device configuration**.

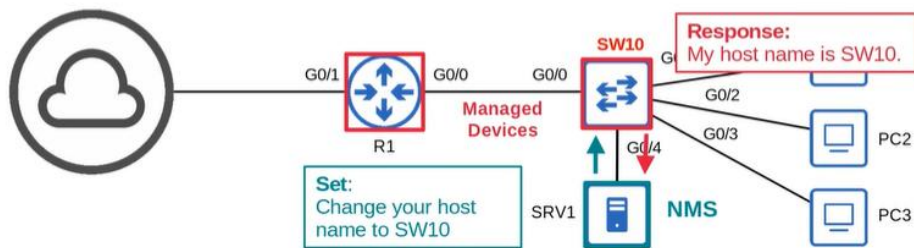
Set

→ A request sent from the manager to the agent to change the value of one or more variables.
The agent will send a *Response* message with the new values.

Real Flow:

NMS → Device: "Change hostname to SW10"

Device → NMS: "Done. Hostname is now SW10"



Explanation:

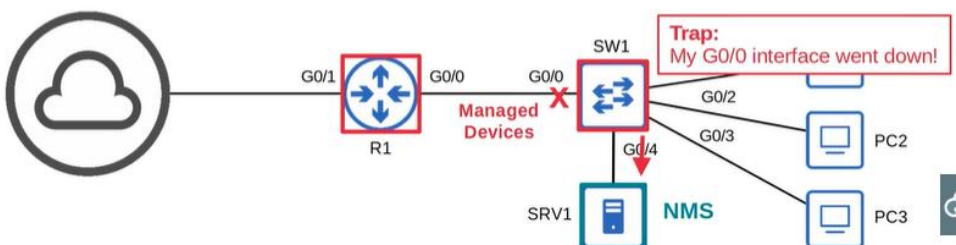
1. Sent from NMS → device
2. Changes values (IP, hostname, etc.)
3. Device replies with **Response**

Notification Messages (Device Alerts NMS)

These messages are sent by **devices** → **NMS** when something happens.

Real Event Example:

SW1 → NMS: "My interface G0/0 went DOWN"



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Types of Notifications

1. Trap

→ A notification sent from the agent to the manager. The manager does not send a Response message to acknowledge that it received the Trap, so these messages are 'unreliable'.

No acknowledgment and sent once

Important Behavior:

1. Uses UDP
2. No retransmission

Meaning: If lost, it is not resent

2. Inform

→ A notification message that is acknowledged with a Response message.
→ Originally used for communications between managers, but later updates allow agents to send Inform messages to managers, too.

Requires acknowledgment and more reliable.

Device → NMS: "Something happened"

NMS → Device: "Received"

Key Difference

Message	Behavior
Trap	No response
Inform	Response required

Response Message (Completing Communication)

Sent as a reply to:

1. Get
2. Set
3. Inform

Simple Flow:

Request → Response

SNMP Agent = UDP 161
SNMP Manager = UDP 162

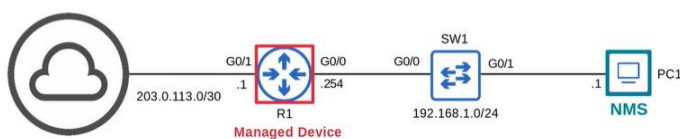
SNMP Configuration

SNMPv2c Configuration

SNMP configuration is not mentioned in the exam topic list.

We not cover setting up the NMS, that definitely out-of-the-scope for the CCNA But let's see how to setup SNMP agent on cisco router

Network Diagram:



R1- Managed Device & PC1 is NMS

CLI Configuration:

```
R1(config)#snmp-server contact jeremy@jeremysitlab.com
R1(config)#snmp-server location Jeremy's House
```

Optional information

Step 1: Optional Information

1. Contact and location
2. Helps identify device
3. Not required

```
R1(config)#snmp-server community Jeremy1 ro
R1(config)#snmp-server community Jeremy2 rw
```

Configure the SNMP *community strings* (passwords)
ro = read only = no Set messages
rw = read/write = can use Set messages

Step 2: Community Strings (Passwords)

Jeremy1 → Read Only (RO) & Jeremy2 → Read/Write (RW)

Meaning:

1. RO → Only Get (read)
2. RW → Get + Set (read + change)

Default Values:

1. public → RO
2. private → RW

Important: Defaults are less secure → should be changed

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```
R1(config)#snmp-server host 192.168.1.1 version 2c Jeremy1
```

Specify the NMS, version, and community

Step 3: Define NMS

NMS IP = 192.168.1.1

Version = v2c

Community = Jeremy1

Meaning:

1. NMS can only read
2. Cannot modify device

```
R1(config)#snmp-server enable traps snmp linkdown linkup  
R1(config)#snmp-server enable traps config
```

Configure the Trap types to send to the NMS

Step 4: Enable Traps

Link-down/ linkup → interface changes

config → configuration changes

Meaning: Device will automatically notify NMS

```
R1(config)#snmp-server contact jeremy@jeremysitlab.com  
R1(config)#snmp-server location Jeremy's House
```

Optional information

```
R1(config)#snmp-server community Jeremy1 ro  
R1(config)#snmp-server community Jeremy2 rw
```

Configure the SNMP community strings (passwords)
ro = read only = no Set messages
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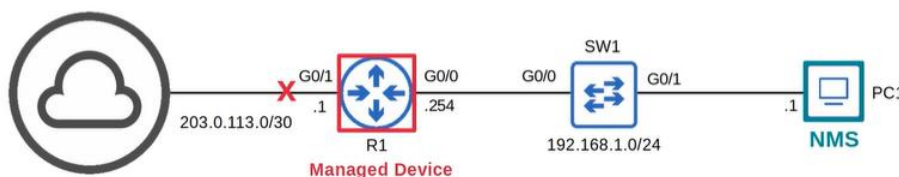
Specify the NMS, version, and community

```
R1(config)#snmp-server enable traps snmp linkdown linkup  
R1(config)#snmp-server enable traps config
```

Configure the Trap types to send to the NMS

So, that's a simple SNMPv2c configuration.

Let's see what happen when R1's G0/1 interface goes down



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Wireshark Capture (Real Packet Understanding)

Here a Wireshark capture of the trap that was sent from R1 to the NMS(PC1)

Packet View:

No.	Time	Source	Destination	Protocol	Length	Info
209	13:55:21.662570	192.168.1.254	192.168.1.1	SNMP	221	snmpV2-trap 1.3.6.1.2.1.

> Frame 209: 221 bytes on wire (1768 bits), 221 bytes captured (1768 bits) on interface -, id 0

> Ethernet II, Src: 0c:1c:1a:87:fb:00 (0c:1c:1a:87:fb:00), Dst: 0c:1c:1a:50:80:01 (0c:1c:1a:50:80:01)

> Internet Protocol Version 4, Src: 192.168.1.254, Dst: 192.168.1.1

> User Datagram Protocol, Src Port: 65385, Dst Port: 162

> Simple Network Management Protocol

version: v2c (1)
community: Jeremy1

> data: snmpV2-trap (7)

snmpV2-trap

request-id: 14
error-status: noError (0)
error-index: 0

variable-bindings: 6 items

> 1.3.6.1.2.1.1.3.0: 104924
> 1.3.6.1.6.3.1.1.4.1.0: 1.3.6.1.6.3.1.1.5.3 (iso.3.6.1.6.3.1.1.5.3)
> 1.3.6.1.2.1.2.2.1.1.2: 2
> 1.3.6.1.2.1.2.2.1.2.2: 4769676162697445746865726e6574302f31
> 1.3.6.1.2.1.2.2.1.3.2: 6
> 1.3.6.1.4.1.9.2.2.1.1.20.2: 61646d696e6973747261746976656c7920646f776e

Protocol: SNMP (UDP)

Source: Device (192.168.1.254)

Destination: NMS (192.168.1.1)

Version: v2c

Community: Jeremy1

Message: Trap

Event: linkDown

Detailed Explanation (Following Packet Fields)

1. Source & Destination

1. Source = Managed device
2. Destination = NMS

→ Confirms Notification flow (device → NMS)

2. Version

1. SNMPv2c used
2. Matches configuration

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3. Community String

Jeremy1 (RO)

Important: → Visible in plain text

4. Message Type

Trap message

Meaning: → Event notification

5. OIDs in Packet

1. Multiple OIDs included
2. One identifies event

Example: → linkDown → interface failure

Critical Security Understanding

SNMPv1 & SNMPv2c:

1. No encryption
2. Data visible

Packets can be captured and read

Why SNMPv3 is Better:

1. Encrypts messages
2. Authenticates devices

